

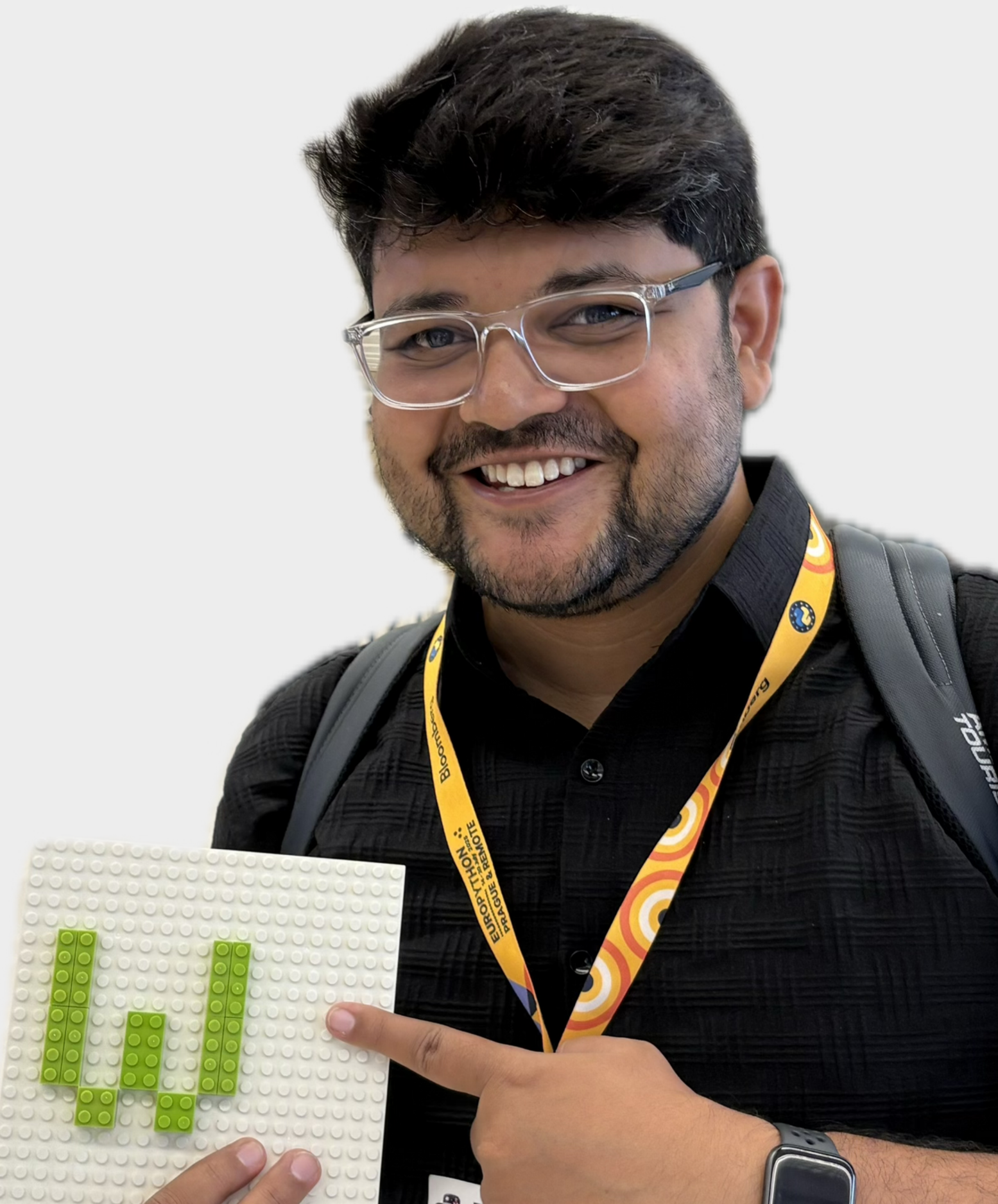
Your Next Server Might Be a Browser

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👋 I'm *Ani*

Author:

- github.com/anistark/wasmrun
- github.com/anistark/feluda
- github.com/anistark/sot

Maintainer:

- github.com/explodinggradients/ragas



The browser today



Mostly seen as a UI layer

Runs JavaScript/Typescript

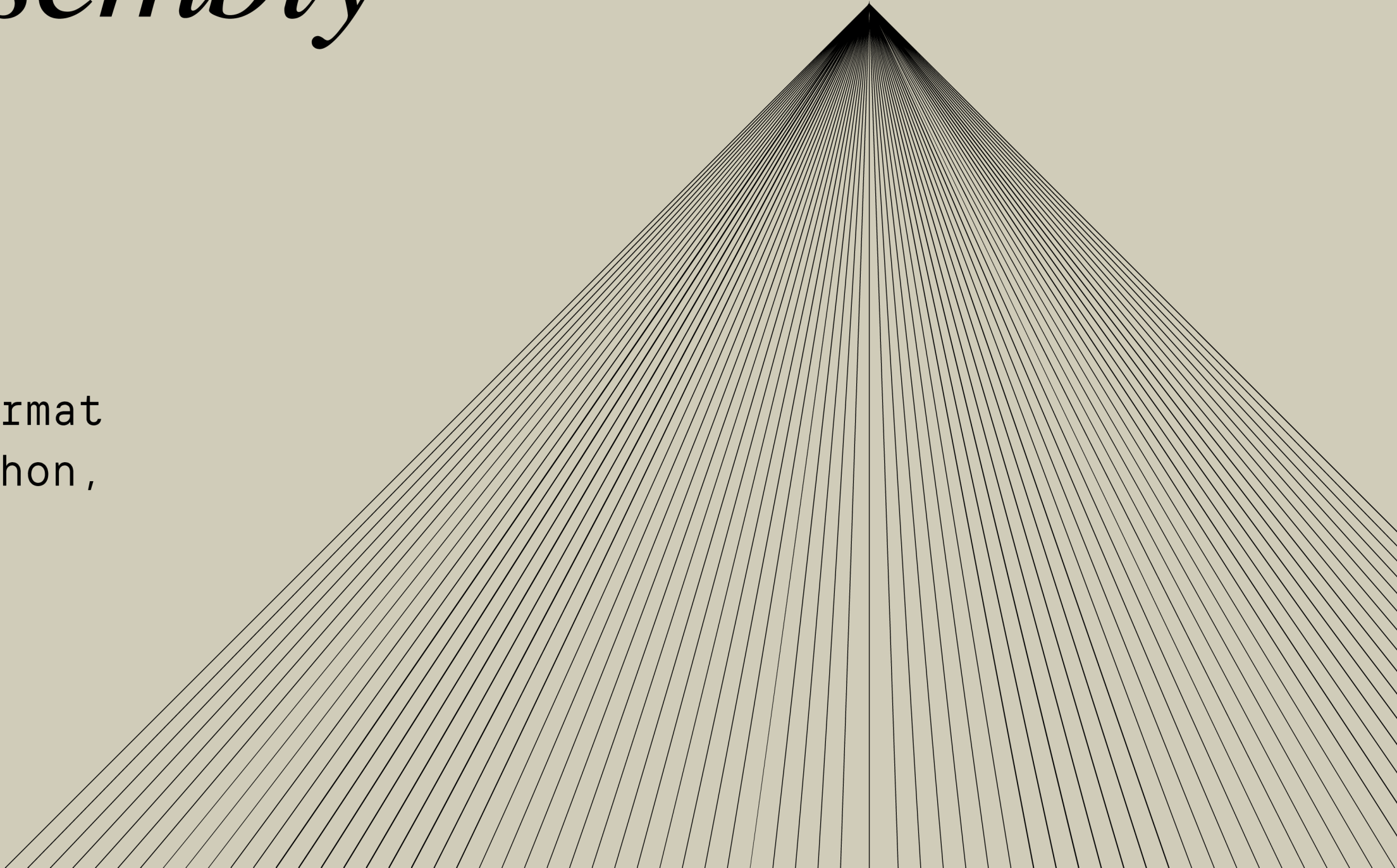


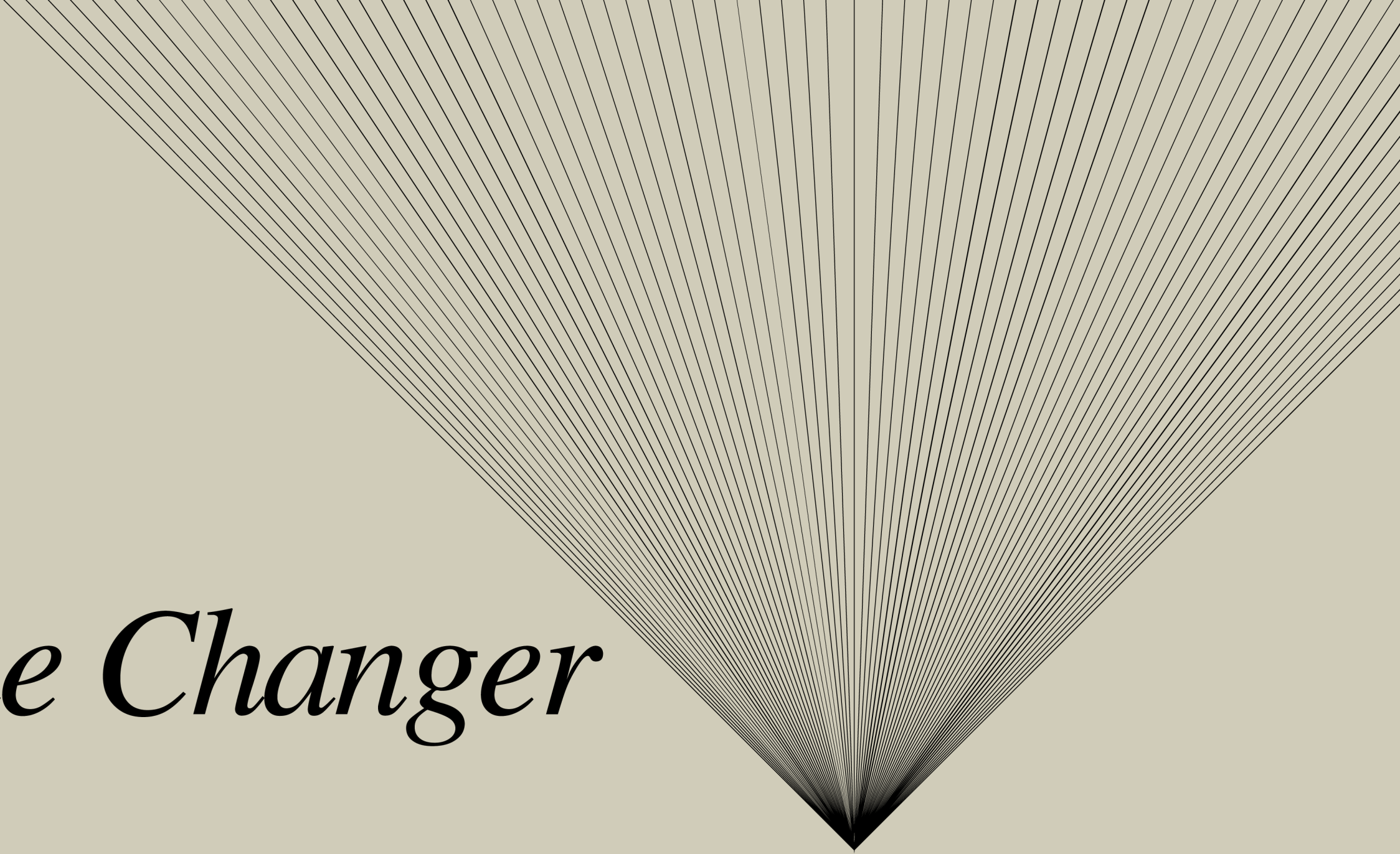
Great for web apps, not for heavy runtimes



Enter WASM

What is WebAssembly (WASM)

- A portable, low-level binary format
 - Runs Rust, Go, C, C++, Zig, Python, TypeScript
 - Near-native performance
 - Safe, sandboxed, small
- 



Why it matters

WASM = Game Changer

- Run code anywhere – browser, edge, server
- No OS dependency
- No heavy containers
- Security by design (sandboxed)

The Possibility *Browser as Runtime*

```
SeaBIOS (version rel-1.16.2-0-gealb7a0)
Booting from DVD/CD...

ISOLINUX 6.04 6.04-pre1 ETCD Copyright (C) 1994-2015 H. Peter Anvin et al
boot:
```

```
  OpenRC 0.54 is starting up Linux 6.6.49-0-virt (i686)
```

```
* /proc is already mounted
* Mounting /run ... [ ok ]
* /run/openrc: creating directory
* /run/lock: creating directory
* /run/lock: correcting owner
* Caching service dependencies ... [ ok ]
* Remounting devtmpfs on /dev ... [ ok ]
* Mounting /dev/mqueue ... [ ok ]
* Mounting modloop ...
```

wasrun

Meet

wasrun

- Open source WASM runtime
- Runs real code inside browsers
- Dev-friendly, portable
- Think “Node for WASM”



```
Loading WASM module: rust_hello_bg.wasm
Analyzing WASM module structure...
Module analysis complete: 13 sections, 63 functions
✅ WASM module loaded successfully!
Found 5 exported functions
```

WASM Outlook

Future for WASM

- WASI (WebAssembly System Interface) maturing
- Browsers becoming self-contained dev environments
- WASM on edge & browsers blurring server/client lines

Key Takeaways

1. Web Dev $\neq$ just JS/TS
2. Containers have lightweight rivals
3. WASM $\neq$ painful (tooling is growing fast)
4. Rethink dev: the browser can be your next VM



Q & A

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